InsideLens

Explainable real-time match intelligence for football

Turn synthetic football events into insights that show their work

GitHub | Azure OpenAI | Translator | SQL | Fabric | Event Hubs

01 PROBLEM

The problem

- Match tools tell you WHAT happened (score, xG, possession) — never WHY.
- Models are black boxes: no evidence, no accountability.
- One feed for everyone ignores analysts, commentators, and fans.
- No audit trail for studios, streamers, or regulators.

02 SOLUTION

Our solution

- InsideLens ingests a live event stream and explains every key moment in real time.
- Every insight ships with 3 explainability pillars: weighted factors, evidence graph, counterfactual.
- Detects tactical patterns: high press, low block, transitions, overloads, sustained pressure, set plays.
- Builds an explainable highlight reel ranked by an excitement score.
- Writes full-time recaps, localized to 6 languages and personalized to 7 personas.

03 DEMO

Live demo

- Pick any 2 of 20 Premier League clubs (2026/27 squads). Default: Man Utd vs Man City.
- 22 players move with the ball; fading trail; action arrows; goal celebrations.
- Streaming WebSocket: match_init -> tick -> insight -> end.
- Expand an insight card: factors (sum to 1), evidence (event ids), counterfactual, translations.
- Switch language and persona live; watch the recap rewrite.

Explainability pillars

- 1. Weighted factors — normalised contributions that sum to 1.
- 2. Evidence graph — human-readable items citing exact event ids and metric deltas.
- 3. Counterfactual — the single change that would most alter the outcome.
- Attached to EVERY insight, not just goals. 12/12 fully explained in eval.
- Confidence score + model provenance on each card.

Mapped to the brief

- Analyze match events in real time — async streaming pipeline over WebSocket.
- Generate stories, highlights, recaps — highlights.py, narrative.py.
- Explain key moments & tactical patterns — explain.py, patterns.py.
- Expand datasets for AI/analytics — 28 engineered features, JSONL/CSV/Parquet/Delta.
- Multiple languages — EN/ES/FR/PT/AR/ZU, online or offline.
- Personalized experiences — 7 personas for fans, studios, broadcasters.

06 STACK

Architecture & Azure

- Live feed: Synthetic / Replay / Azure Event Hubs (pluggable, lazy).
- Analytics: possession, xG, momentum, field tilt, press intensity, flanks.
- Narratives: Azure OpenAI on Microsoft Foundry (local template fallback).
- Translation: Azure AI Translator (offline template fallback).
- Persistence: Azure SQL Database; analytics lakehouse: Microsoft Fabric / OneLake.
- Runs with ZERO cloud keys; any Azure failure auto-falls back. Demo never breaks.

07 DATA

Dataset expansion

- 28-column ML-ready export: zone, flank, distance-to-goal, shot angle, pressure score.
- Formats: JSONL, CSV, Parquet, Delta (deltalake + pyarrow).
- Upload straight to Fabric OneLake with upload=true.
- Synthetic, football-realistic data only — no real Premier League match data.
- Eval harness: 12 insights, 100% explained, factor sums = 1.0.

08 REACH

Personalization & multilingual

- 7 personas: studio-analyst, global-fan-es/fr/pt/ar, fan-home, fan-away.
- Tone, depth, stat-bias, favourite team, language per persona.
- Per-insight language chips; recap rewrites per persona.
- Conversational analyst endpoint /api/agent — tool-grounded, multilingual.
- Spatial viz: heatmaps, pass networks, expected threat (xT).

09 QUALITY

Proof

- 78 tests passing, 5 skipped offline; CI on Python 3.11 & 3.12.
- Azure smoke exercise reports PASS / FALLBACK / SKIP per provider.
- Deterministic seeded simulator (INSIDELENS_MATCH_SEED) — reproducible demos.
- Lazy Azure imports — no hard dependency in requirements.txt.
- GitHub Copilot instructions + AGENTS.md for AI-assisted development.

10 NEXT

Roadmap & ask

- Host on Azure Container Apps; live demo URL for judges.
- Add autonomous Foundry agent with tool-calling over match state.
- Voice commentary + live broadcast graphics overlay.
- Pilot with a studio: real-time insight feed into production.
- Ask: feedback on explainability UX and Azure deployment topology.

11 CONTACT

Thank you

InsideLens

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Explainable real-time match intelligence | Azure-ready, offline-first